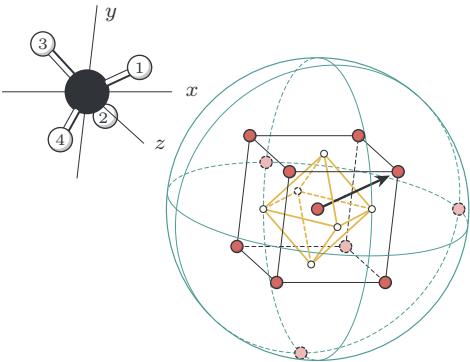
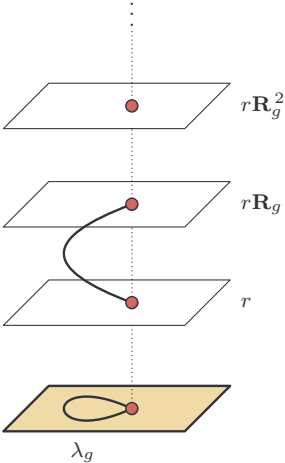


methane, CH₄: the reference **X**₀ in its body frame,
the *C*₂ axes *x*, *y*, *z* through the midpoints of opposite edges



the orientation space *SO*(3) as a ball,
direction the axis and distance the angle:
the twelve rotations of *T* in red,
the cell *U* at *q* = 0 in gold,
and the lift *r* → *rR*_{*g*} of the
third-turn about bond 1



over the base *SO*(3)/*T*, twelve sheets,
one for each rotation in *T*, three drawn;
the third-turn closes as a loop *λ*_{*g*} in the base
and, lifted, climbs from sheet *r* to sheet *rR*_{*g*};
A sees the climb as 1, ¹E as *e*^{2*πi*/3},
²E as *e*^{4*πi*/3}, T as *x* → *y* → *z*